

SmartCare v0.4 - Domain Implementation Workbook

Week 7 student resource

1. UML-to-Code Trace

UML element	Python element	Implemented?	Notes
Patient class	Class patient	Yes	Matches approved patient class
Patient ID	Patient_id	Yes	Stored as instance attribute
Patient Name	Self.name	Yes	Stored as instance attribute
Create record	Patient.create_record	Yes	Creates/displays patient record
Search by name	Search_by_name	Yes	Allows to be filtered by names
Update details	Update_details()	Yes	Updates the patients details
Retrieve information	Retrieve_information	Yes	Fetches the patients details
Practitioner Class	Class Practitioner	Yes	Matches approved practitioner class
Practitioner ID	Practitioner_id	Yes	Stored as an instance attribute
Practitioner Name	Self.name	Yes	Stored as an instance attribute
Practitioner specialty	Self.specialty	Yes	Stored as an instance attribute
Practitioner Availability	Self.availability	Yes	Stored as an instance attribute
View Schedule	View_schedule()	Yes	Returns current schedule
View Availability	View_availability()	Yes	Returns available times
Update Availability	Update_availability()	Yes	Updates available times

2. Domain Invariants

Class	Invariant / rule	How protected
Patient	Patient ID must be positive	__init__() raises ValueError if patient_id <= 0.
Patient	Patient name cannot be empty	__init__() checks name.strip()
Practitioner	Practitioner ID must be positive	__init__() raises ValueError if practitioner_id <= 0.
Practitioner	Practitioner name cannot be empty	__init__() checks name.strip()

3. Composition / Inheritance Decisions

Relationship	Decision	Rationale
Patient → Practitioner	Association	Patients and practitioners interact via appointments rather a direct relationship.
Schedule → Practitioner	Rejected	Requirements mention schedules and availability but do not provide enough evidence for an independent Schedule class.
Patient → Practitioner inheritance	Rejected	Patient and practitioner types are separate entities and neither are specialised type of the other.

4. AI Pair-Programming Record

AI contribution	Conforms?	Decision	Reason	Verification
Generated AppointmentStatus enum	Yes	Reject	Not asked within the workbook	This is an unsupported class
Generated Appointment class and attributes	Yes	Reject	Not asked within the workbook	This is an unsupported class
Added input validation	Yes	Keep	Supports basic validation requirements without adding unsupported functionality.	Tested invalid appointment ID, empty date/time, missing patient and missing practitioner.
Added cancel() method	Yes	Keep	Match the approved UML rule and cancellation rule	Created an appointment and confirmed its status changed to CANCELLED.
Added update_status()	Yes	Keep	Matched the approved UML and protects status transitions	Tested valid status input.
Added release_booking_slot()	Yes	Keep	Matches the approved UML and ensures cancellation triggers slot release behaviour.	Confirmed cancel() calls release_booking_slot().

5. Updated UML

Insert updated UML only if implementation revealed a justified design change. Explain every change.

No changes made